MILITARY SPECIFICATION

SIGHT, BORE, OPTICAL: M45E1

1. SCOPE

1.1 This specification covers one type of instrument known as Sight, Bore, Optical: M45E1 used for boresighting mortars. Throughout this specification this instrument is referenced as "BOre sight"- Telescope, Elbow Assembly 10548042-2 Is an integral component of this bore sight.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	parts, Equipment and TOOLS for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Supply and Maintenance of
MIL-T-46324	Telescope, Elbow, M109 and Telescope, Elbow Assembly 10548042-2.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F10548286	Sight, Bore, Optical : M45E1

PACKAGING DATA SHEET

10549221	Packaging of Sight, Bore, optical: M45E1 W/E

FSC 1240

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(Copies of specifications, standards, drawings and packing data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The bore sight shall be manufactured in accordance with Drawing F10548286.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Telescope, Elbow, AssemblY 10548O42-2.- The Elbow Telescopes which is a part of the bore sight, shall have met the requirements of MIL-T-46324 prior to assembly as a bore sight.

3.5 Orientation.- The bore sight shall be clamped securely in each of the three positions shown in Figure 1 with the respective V slide contacting the longitudinal portion of a smooth 4.5 inch diameter tube or equivalent, whose axis is inclined 45 degrees above the horizontal. The following conditions shall be met at each of the above specified positions unless otherwise specified.

3.6 Accuracy.

3.6.1 Collimation.- With the cross level vial bubble centered within 1/2 the space width between two graduation lines, the line of sight of the elbow telescope shall be parallel to the vertical plane passing through the axis of the corresponding V slide within 3 roils for positions (a) and (c) of Figure 1.

3.6.2 Plumb travel.- With the cross level vial bubble centered within 1/2 the space width between two graduation lines, and using position (a) of Figure 1 in accordance with 3.5, with the line of sight of the elbow telescope in a horizontal plane and in coincidence with a plumb line; the line of sight shall not deviate from the plumb line by more than 2.0 mils, (4.0 mils total displacement) from maximum depression to maximum elevation.

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3.6.3 Inverted level vial.- With the level vial bubble centered within 1/2 the space width between two graduation lines, using position (b) of Figure 1, the line of sight shall not deviate from the plumb line by more than 1.0 mil (2.0 roils total spread) when elevated 30 degrees above the horizontal.

3.6.4 Elevation level vial.- With the cross level vial bubble centered within 1/2 the space width between two graduation lines, the elevation level vial bubble shall be centered within 1/2 the space width between two graduation lines.

3.7 Telescope travel.- With position (a) of Figure 1 established in accordance with 3.5, the telescope shall be capable of being rotated a minimum of 45 degrees in elevation and 30 degrees in depression.

3.8 Operating temperature.- The bore sight shall meet the torque requirements of 3.9 when thermally stabilized at ambient temperatures of minus 40 degrees Fahrenheit ad-plus 150 degrees Fahrenheit and the strap assembly shall be completely operable when securing and loosening the bore sight to the tube.

3.8.1 Clamp screw. - The clamp screw, in a free state, shall be operable throughout its threaded portion without restriction or binding at all temperatures specified below.

3.9 Torque.- The torque required to elevate and depress the telescope through full travel shall not exceed 10 pound inches nor be less than 2 pound inches at room temperature (60 to 90°F). The torque required to elevate and depress the telescope through full travel shall not exceed 28 pound inches nor be less than 2 pound inches at the extreme temperatures specified in 3.8.

3.10 Strap assembly.- The bore sight or strap assembly shall not become loosened as a result of applying 2 pounds direct force at the knurled portion of the clamping screw, once in each opposing direction parallel to the tube axis, when the bore sight is positioned and secured as shown in (a) of Figure 1 with a torque of 2 ± 1/4 pound inches applied to the clamp screw.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirement of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 bore sights plus 3 each of all items covered by sQAP. The sample shall be manufactured in

ar production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Governemnt is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, ‘including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection Provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shell be inspected in accordance with the Inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP' the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 0.65% Defective			
101.	Collimation	3.6.1	4.6.2
102.	Inverted level vial	3.6.3	4.6.4
103.	Elevation level vial	3.6.4	4.6.5
104.	Telescope travel	3.7	4.6.6
105.	Clamp screw (60° to 90°F)	3.8.1	4.6.8
106.	Torque (60° to 90°F)	3.9	4.6.9
107.	Strap assembly	3.10	4.6.10
108.	Plumb travel	3.6.2	4.6.3
MINOR: None Defined			

NOTE: The tests in Table I shall be conducted at a temperature between +60 and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and readmittance of rejected lots shall be in accordance with “Acceptance and Rejection" as specified in MXL-STD-105.

4.3.4 Certification.- Certification for Telescope, Elbow Assembly, 10548042-2 shall include test data and results for each requirement specified in MIL-T-46324 to insure compliance with 3.4. Certification shall be provided prior to performing final acceptance inspection for the bore sight and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

4.4 Special sampling.

4.4.1 General.- Three bore sights shall be selected at random as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
301.	Materials	3.1	Applicable Drawings - visual
302.	Fabrication	3.2	Applicable Drawings - Visual
303 l	General specification	3.3	MIL-F-13926 - Visual

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4.4.2 Environmental.- Three bore sights shall be selected at random as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
304.	Operating temperature	3.8	4.6.7
305.	Clamp screw	3.8.1	4.6.8
306.	Torque	3.9	4.6.9

NOTE: The tests in Table III shall be conducted at temperatures of minus 40 degrees Fahrenheit and plus 150 degrees Fahrenheit.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does

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not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to favilitate routine calibration of test equipment.

4.5.2.2.1 Orientation.- The contractor shall provide special inspection equipment in accordance with 3.6.1 and 4.6.1.

4.6 Test methods and procedures.

4.6.1 Orientation.- Orientation is accomplished by means of a special testing device to assure correct orientation in the three positions indicated on Figure 1. The device shall consist of a slimated tube, with a base capable of supporting the tube whose axis is inclined 45 degrees plus or minus 20 seconds, above the horizontal. The outside diameter of the tube shall measure 4.500 inches plus .050 inch and parallelism shall be within .003 inch per foot. The surface quality shall be 63 micro inches RMS.

4.6.2 Collimation.- This test is accomplished by means of two plumb lines located approximately fifty feet apart, and the tube of the device positioned midway between the plumb lines so that the axis of the tube lies in the plane of the plumb lines. This can be checked by locating a vertical marker on the center line of the tube at each end. By means of a collimator, located approximately 15 feet from the back of the plumb line, bring the two plumb lines and the two markers on the tube in alinement with the vertical lines of the collimator. A peep sight placed on the objective end of the collimator, with a 3/16 inch aperture will enable the observer to make an accurate reading. Mount the bare sight on the tube, as illustrated in (a) of Figure 1, by means of the strap assembly. Rotate the bore sight around the outside diameter of the tube until the cross level bubble is centered, and tighten the clamp screw. Veiw through the telescope the image of the plumb line. The vertical line of the telescope shall coincide with the plumb line within the tolerance indicated in 3.6.1 as read on a mil chart placed in back of the plumb line. The bore sight is then reversed on the tube, as illustrated in (c) of Figure 1, and the same cross level and viewing procedure employed, using the opposite plumb line.

4.6.3 Plumb travel.- Plumb travel check consists of mounting the bore sight on the tube by means of the strap assembly as shown in position (a) of Figure 1. Locate a plumb line as described in 4.6.4. Center cross level vial bubble within 1/2 the space width between two graduation lines. The plumb line as viewed through the telescope at the intersection of the cross lines of the reticle shall not deviate more than the tolerance of 3.6.2 when the telescope is displaced from maximum elevation to maximum depression. A mil block chart, located at one end of the plumb line, will indicate the amount of deviation.

4.6.4 Inverted level vial.- Inverted level vial check for accuracy consists of mounting the bore sight on the tube of the device by means of

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the strap assembly in the position (b) of Figure 1. Locate a plumb line precisely eight feet from the center of the pivot shaft of the bore sight and in a vertical plane with the axis of the tube. By means of a peep sight with a 1/8 inch aperture placed on the objective end of the telescope, move the bore sight until the cross mark of the telescope coincides with the plumb line when elevated 30 degrees above the horizontal. In this position, check the level vial which shall meet the requirenmts of 3.6.3.

4.6.5 Elevation level vial.- Elevation level vial test is made by checking the 45 degree angle of the tube of the device by means of a commercial optical protractor. Mount the bare sight on the tube as illustrated in (a) of Figure 1, and cross level. In this position, the elevation level vial shall be central within the tolerance of 3.6,4.The same test is repeated for checking the other "V" slide with the bore sight mounted as illustrated in (c) of Figure 1.

4.6.6 Telescope travel.- Telescope travel is accomplished with the bore sight and plumb line located as indicated in 4.6.4. Locate a mark on a chart eight feet above the horizontal line of sight which will indicate 45 degree elevation point, and another mark feet 7-1/2 inches below the horizontal line of sight which will indicate the 30 degree depression point.The telescope shall travel from one point to the other to meet requirements of 3.7.

4.6.7 Operating temperature.- Operating temperature test is made by clamping the bore sight to a 4.50 inch outside diameter steel tube by means of the strap assembly and exposing it to temperatures indicated in 3.8 by means of a commercial cold chest and oven. When thermally stabilized at these temperatures, the torque requirement of 3.9 shall be made by means of a commercial spring scale attached to the end of the telescope housing. The operability of the strap assembly is then checked by releasing the clamp and moving the bore sight, and re-clamping at different positions.

4.6.8 Clamp screw.- Clamp screw check is a manual test made by revolving the screw from one end of the thread to the other end without binding or other restrictions at all temperatures herein specified.

4.6.9 Torque.- Torque test shall be made by means of a spring scale which can be hooked at one end of the telescope housing. The telescope is then moved through its full travel by pulling the scale and observing the indicator of the scale for requirements Of 3.9.

4.6.10 Strap assembly.- Strap assembly test is performed by mounting the bore sight on the tube of the device in (a) of Figure 1, and by means of a torque wrench and adapter, apply a pressure of two pound inches to the clamp screw. Sight the telescope on a target at the intersection of the crosslines of the telescope. Hook a spring scale on the knurled portion of the clamp screw and apply the pressure indicated in 3.10 in opposing directions parallel to the tube axis. Observe the target thru the

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telescope and there shall be no relative movement of the telescope.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10549221. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- In preparation for firing mounted or dismounted, the bore sight is used to establish a definite line of sight when boresighting Mortar, 81mm Series, Mortar, 4.2 inch series and Mortar, Self-propelled, Full Track, 107mm.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation packaging and packing."
(c) Applicable stock number.
(d) Provisions for First Article testing.

Custodian:

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Preparing activity:

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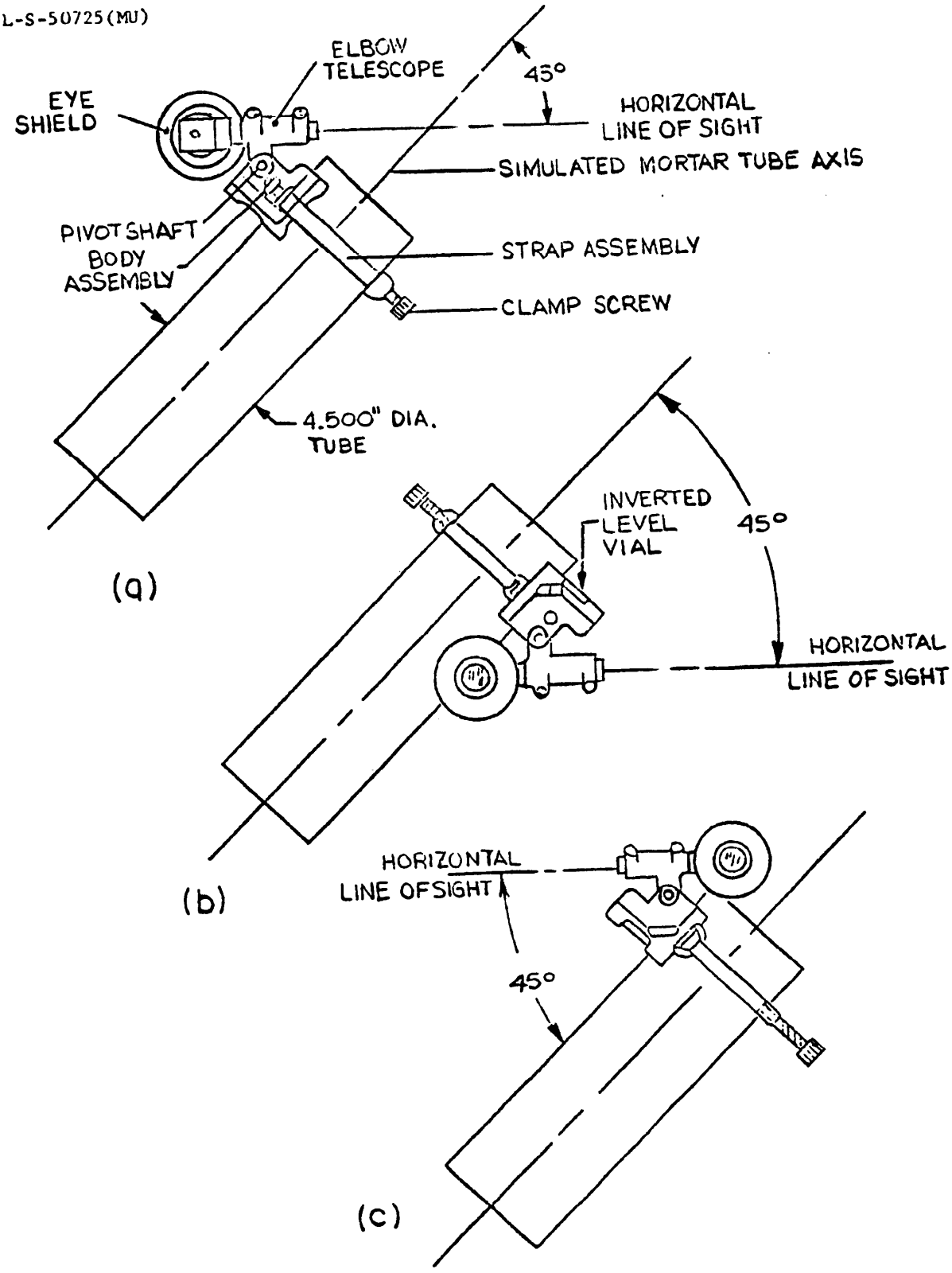


FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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